

SHANGZE KONG

BEng Mechanical Engineering,
UCL
2024–2027 · First Class
expected

**Personal Portfolio · Robotics
/ Embodied AI**

From vision-language perception to embodied action on humanoid platforms

A VLA / embodied-AI research path connecting multimodal evidence diagnosis, real humanoid calibration, and closed-loop robotic action.

A research path converging on VLA and embodied AI

The common question: how vision-language perception grounds reliable embodied action.

University College London

BEng Mechanical Engineering · 2024–2027 · First Class expected · 71.84%



Research identity

I study how vision-language perception can be grounded into reliable embodied action — especially on humanoid platforms where perception errors become physical risk.

71.84%

UCL average score
First Class expected

ACM MM

2026 Accepted
First Author · CCF-A

2 Papers

1 published
1 accepted

3 Labs

Research positions
Fudan · Cambridge · UCL

1 Exhibition

Humanoid robot showcase
CISMA

VLA coverage across my projects

VL front-end

PercepAgent: evidence-gap diagnosis and tool-augmented VLM agent

Action end

Fudan humanoid: perception→registration→closed-loop grasp

Embodied breadth

Cambridge: bio-inspired navigation, sensor fusion, VLA survey

V/L capability: diagnosing missing evidence in multimodal models

Accepted at ACM MM 2026 (First Author) · the vision-language front-end of a VLA system.

Research question

When VQA fails, is the bottleneck really reasoning / policy capacity — or missing visual evidence?

Tool-assisted oracle ceiling

Per sample, choose the best output among frozen baseline and four tool-augmented variants. This separates information-access failure from capacity failure.

VLA hook

Before blaming a policy, diagnose whether the V/L front-end actually sees the evidence the action needs.

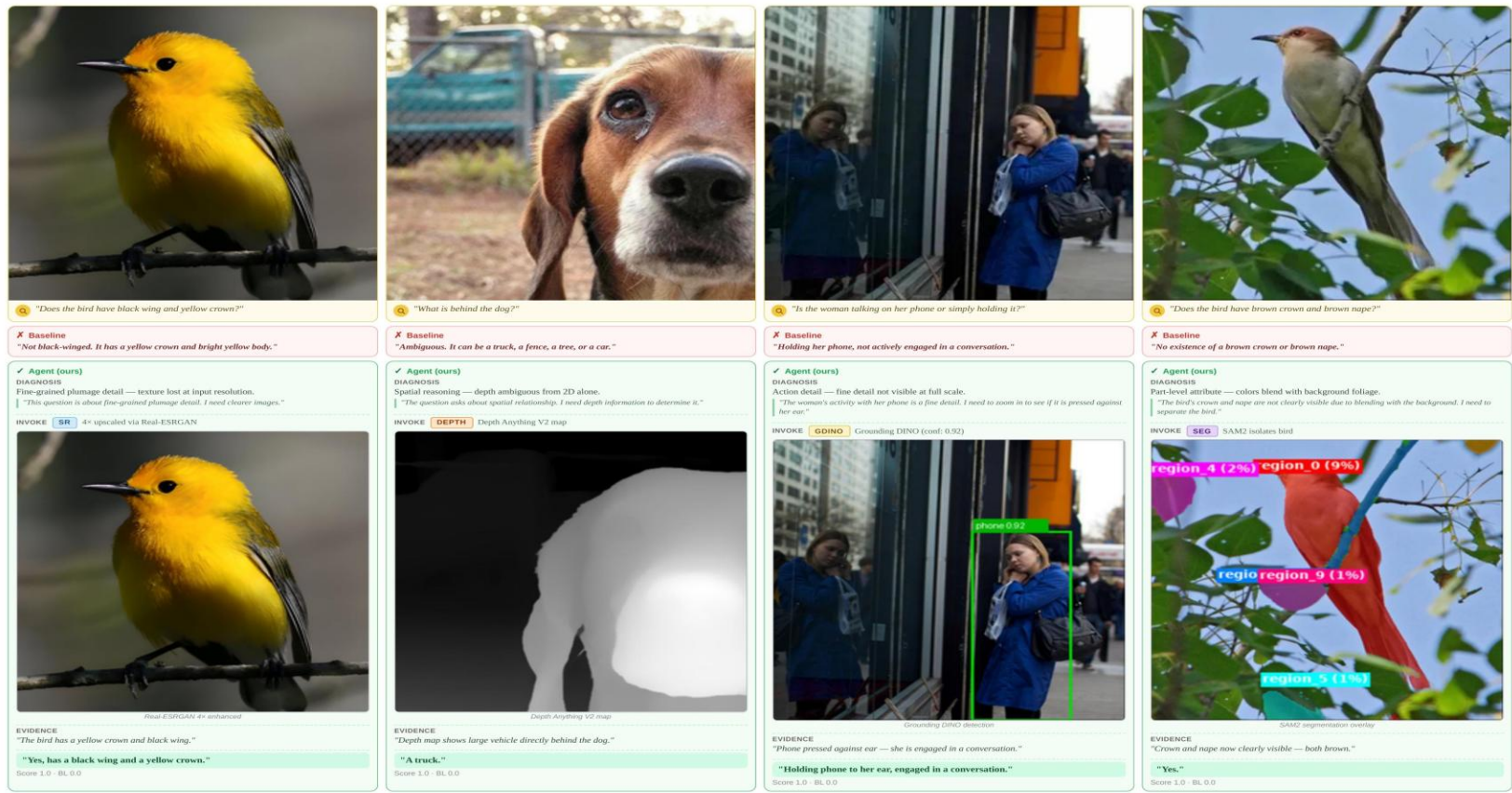


Figure 1. VLMs fail on visually grounded tasks not due to reasoning, but due to perceptual information gaps. Our training-free agent diagnoses the failure type (with explicit chain-of-thought reasoning), then invokes the appropriate perception tool — super-resolution (SR), depth estimation (Depth), grounding detection (GDino), or segmentation (Seg) — to recover the missing visual evidence, consistently correcting baseline errors across four distinct perception gap categories.

Qualitative examples: SR, depth, grounding and segmentation recover different kinds of evidence gaps.

The perception ceiling is large, training-free and transferable

Accepted at ACM MM 2026 (First Author)

Five-benchmark results

Method	Train	Tools	CUB	Flickr	GQA	POPE	VSR	Avg.	Δ_{BL}
<i>Qwen2-VL-7B</i>									
Baseline	X	0	.558 $\pm$.043	.634 $\pm$.032	.608 $\pm$.042	.867 $\pm$.033	.640 $\pm$.043	.661 $\pm$.018	—
VPT (GD) [23] †	522k	1	.930 $\pm$.022	.720 $\pm$.030	.590 $\pm$.042	.883 $\pm$.031	.730 $\pm$.040	.771 $\pm$.016	+16.6%
VPT (Full) [23] †	522k	2	.921 $\pm$.024	.680 $\pm$.031	.623 $\pm$.042	.903 $\pm$.029	.788 $\pm$.036	.783 $\pm$.016	+18.5%
PercepAgent (ours)	X	4	.800 $\pm$.035	.740 $\pm$.029	.673 $\pm$.041	.940 $\pm$.023	.710 $\pm$.040	.773 $\pm$.016	+16.9%
Oracle ‡	X	4	.919 $\pm$.024	.790 $\pm$.027	.753 $\pm$.037	.945 $\pm$.022	.817 $\pm$.034	.845 $\pm$.014	—
<i>Qwen3-VL-8B</i>									
Baseline	X	0	.720 $\pm$.039	.814 $\pm$.026	.520 $\pm$.043	.900 $\pm$.029	.687 $\pm$.041	.728 $\pm$.017	—
PercepAgent (ours)	X	4	.850 $\pm$.031	.820 $\pm$.026	.710 $\pm$.039	.950 $\pm$.021	.700 $\pm$.041	.806 $\pm$.015	+10.7%
Oracle ‡	X	4	.940 $\pm$.021	.850 $\pm$.024	.743 $\pm$.038	.940 $\pm$.023	.852 $\pm$.032	.865 $\pm$.013	—
<i>Single-tool variants (Qwen2-VL-7B)</i>									
+ SR only	X	1	.678 $\pm$.041	.694 $\pm$.031	.635 $\pm$.042	.905 $\pm$.028	.672 $\pm$.042	.717 $\pm$.017	+8.5%
+ GD only	X	1	.597 $\pm$.043	.565 $\pm$.033	.618 $\pm$.042	.919 $\pm$.026	.648 $\pm$.043	.669 $\pm$.018	+1.2%
+ DE only	X	1	.697 $\pm$.040	.658 $\pm$.032	.624 $\pm$.042	.913 $\pm$.027	.691 $\pm$.041	.717 $\pm$.017	+8.3%
+ Seg only	X	1	.756 $\pm$.037	.649 $\pm$.032	.651 $\pm$.041	.916 $\pm$.027	.680 $\pm$.042	.731 $\pm$.017	+10.6%
<i>Tool-selection strategy (Qwen2-VL-7B)</i>									
Random-tool	X	4	.740 $\pm$.038	.698 $\pm$.031	.560 $\pm$.043	.900 $\pm$.029	.676 $\pm$.042	.715 $\pm$.017	+8.2%
All-tools	X	4	.760 $\pm$.037	.750 $\pm$.029	.600 $\pm$.042	.910 $\pm$.028	.592 $\pm$.044	.722 $\pm$.017	+9.2%
ReActAgent [22]	X	4	.690 $\pm$.040	.591 $\pm$.032	.610 $\pm$.041	.840 $\pm$.035	.580 $\pm$.044	.662 $\pm$.018	+0.1%

.661 → .773

(+16.9% rel.)
PercepAgent vs Qwen2
baseline

.661 → .845

(+17.4 pp abs.)
Oracle vs Qwen2 baseline

61%

Oracle headroom
recovered

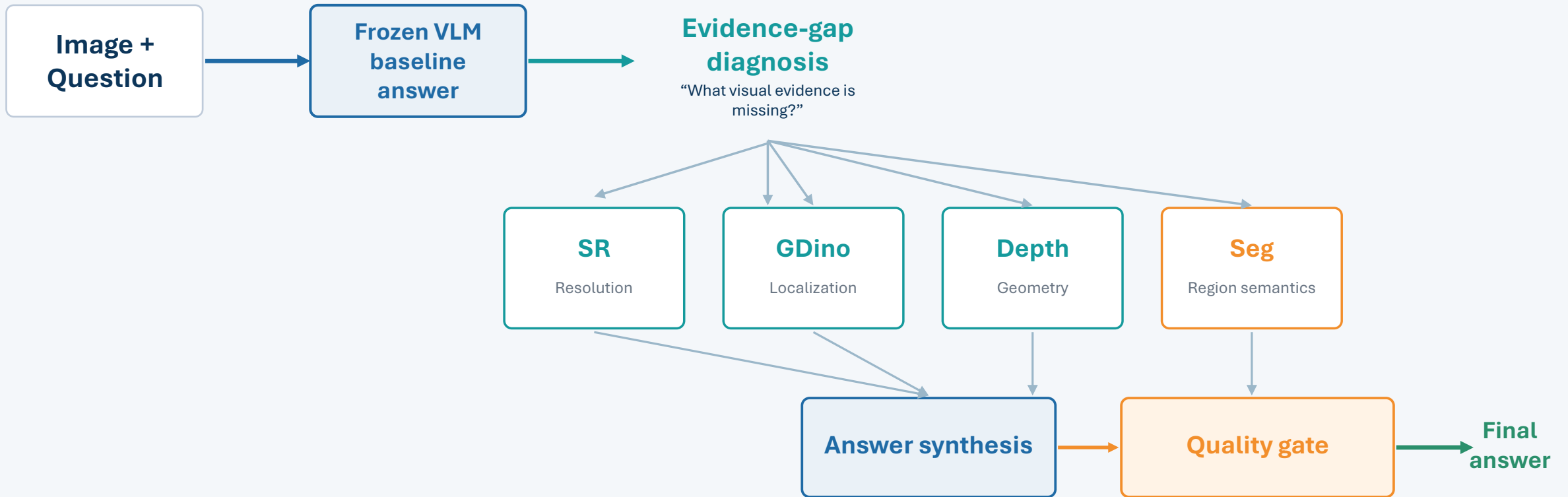
.806

Qwen3-VL-8B
zero reconfiguration

**Oracle .845 > VPT .783 trained on 522k
samples**

Cross-backbone transfer: Qwen3-VL-8B reaches .806 (+10.7%)
with the same selector, prompts and thresholds.

A training-free agent that asks what evidence is missing



Key design choice

Evidence-gap diagnosis, not confidence routing. ReAct-style confidence routing improved only +0.1 pp.

VLA implication

Humanoid manipulation needs object location, distance, precise boundaries and fine details before contact.

Safety lower bound

Baseline-First quality gate: if tool evidence is unreliable, fall back to the frozen baseline.

Baseline-First is the V/L analogue of safe fallback: do not commit uncertain perception when it may drive physical action.

Action side: real humanoid perception→control deployment

Research Assistant, Zhongjie Group project, Fudan University.

Project context

¥25M-level industry project with Zhongjie Group for humanoid robot template loading and unloading in garment manufacturing.

Constraints

- TCP calibration unavailable
- AGV docking error $>1\text{ mm}$ / $>1^\circ$
- Contact-rich template grasping

Technical response

External virtual TCP calibration chain: OpenCV solvePnP, hand-eye chaining, least squares, RANSAC and frame transforms.

Humanoid relevance

A real perception→action loop that a VLA policy ultimately has to drive.

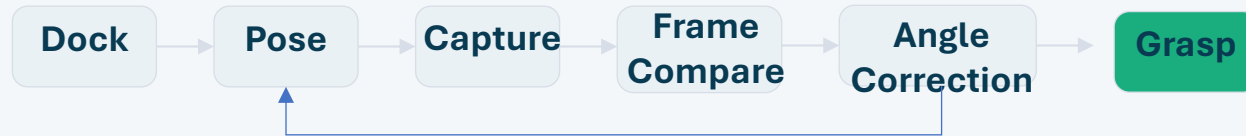


Real-robot platform and deployment context: mobile manipulator prototypes and final humanoid platform.

From observability to closed-loop humanoid manipulation

The physical half of VLA: perception must become calibrated, observable and executable action.

State machine loop



- Docking
- Vision
- Registration
- End-effector

± 0.5 mm
Grasping accuracy

$< 0.5^\circ$
Rotation error after
correction

ROS/MoveIt
Integrated perception &
grasping

CISMA
System exhibited

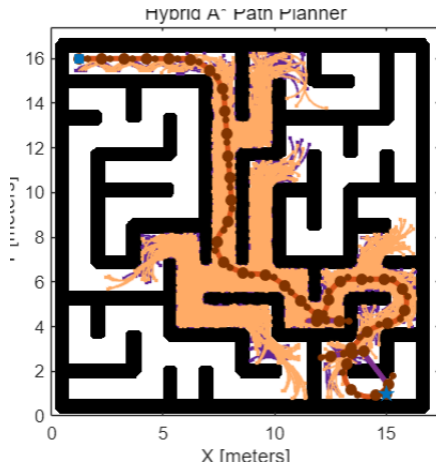
Engineering method

Observability first $\rightarrow$ error budget $\rightarrow$ design $\rightarrow$ measure $\rightarrow$ analyze $\rightarrow$ iterate. This is how I think about precise contact-rich humanoid manipulation.

For humanoid robotics, this is the physical half of VLA: perception must become calibrated, observable and executable action.

Embodied navigation, bio-inspired robotics and multi-sensor fusion

Project Lead · University of Cambridge online research project.



Maze navigation simulation:
occupancy grid, deadlock recovery
and one-step MPC.

Navigation algorithm

- Binary occupancy grid and ray-casting sensing
- Three-pronged cycle detection
- Direction switching, hysteresis, forbidden zones
- One-step MPC centering correction

Humanoid link

Bio-inspired embodied intelligence
+ planning side of VLA task
execution.

Published first-author review

Multi-Sensor Fusion and Collaborative Perception for
Autonomous Robots in Complex Maze Environments
Applied and Computational Engineering, 191(1), 75–84

data / feature / decision / hybrid; KF-family, PF-family
and BEV algorithms

Surveyed 52 papers (including CVPR / ICLR 2025–26 VLA
models) to map architectures, data strategies and open
problems.

VLA = vision-language perception + embodied action

My projects cover the V/L front-end, the action end, and embodied navigation / fusion breadth.

61%

Oracle headroom recovered

VL evidence diagnosis

± 0.5 mm

Humanoid grasping accuracy

Action / control on real hardware

2 Papers

1 published[1]

1 accepted[2]

I work toward vision-language-action models for humanoid perception and control.

From seeing the right evidence → to choosing the right action → to recovering safely when evidence is incomplete.

One-sentence positioning: before a humanoid acts, it should know what it saw, what it missed, and how much to trust that evidence.